

CEP Mapping Form for Project Submission

Course: CSE 414

Project Title: Fake Audio Detection Using LSTM

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CEP Outcome Mapping

CEP Code	Description	Evidence in Project	Student Remarks
K2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline	The project uses MFCC to extract features from the raw audio data, which is a mathematical technique that uses Fourier transforms and Mel-scale filterbanks to convert raw audio into numerical feature vectors. Padding the sequences is to ensure the model gets the same input dimension. The sequences were padded to the same dimension. The performance of three models (LSTM, CNN, GRU) was statistically evaluated and compared using accuracy, precision, recall and F1-score.	
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline	The project employs deep learning knowledge such as LSTM gating mechanisms, CNN convolutional blocks and GRU recurrent units which are all established architectures in the audio classification field. Sequential	

		audio classification was performed with a stacked LSTM (64 to 32 units) with Masking and Dropout (0.2), with forget, input and output gates. We constructed a 1D CNN consisting of two Conv1D layers (64 to 128 filters) and MaxPooling1D to extract local features of MFCC feature sequences. We used a stacked GRU (64 to 32 units) with reset and update gates as a computationally efficient recurrent alternative to LSTM.	
K8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues	Deepfake audio detection is an important and emerging research problem. Three recent research papers were reviewed and cited including works published in Scientific Reports (2025), Circuits Systems and Signal Processing (2026) and an IEEE AIIoT conference (2024). The study critically evaluates if LSTM, CNN or GRU provides the best trade-off between accuracy and efficiency for this emerging security challenge.	
P1	Depth of Knowledge Required	The project requires a high level of technical expertise in different related areas like signal processing	

		(MFCC extraction), deep learning architectures (LSTM, CNN, GRU), data augmentation methods, sequence padding and binary classification.	
P7	Interdependence	The project integrates knowledge from multiple areas digital signal processing, machine learning, cybersecurity, and software engineering to build a complete detection pipeline from raw audio to classification output. The work connects audio forensics with practical security applications.	
A3	Innovation	We used a new data augmentation method with MFCC amplitude scaling (factors 1.05 and 0.95) to increase the size of the training set. We benchmarked three architectures under the same conditions to identify the most effective approach, providing new empirical evidence to the deepfake audio detection literature.	

Detailed Justification

CEP Code	Brief Explanation of How the Project Demonstrates This Outcome
K2	The project implements Mel-frequency cepstral coefficient (MFCC) extraction which is a mathematically based method that uses Fourier transforms, Mel-scale filterbanks and discrete cosine transforms to convert raw audio signals into numerical feature matrices. Model performance is evaluated with formal statistical metrics including precision, recall, F1-score and accuracy based on confusion matrix analysis, illustrating the use of mathematics and numerical analysis for the support of modeling within the discipline.
K4	The project builds on existing specialized knowledge in deep learning for sequential data. Accepted practices in the field are followed. LSTM networks are implemented with input, forget and output gating mechanisms. CNN architectures with Conv1D and MaxPooling layers are implemented. GRU networks with reset and update gates are implemented. The use of dropout regularization, Adam optimization and categorical cross-entropy loss is based on theoretical frameworks from the present state of the art in the field of neural network engineering.
K8	The related work section reviews three recent publications from 2024 to 2026 in the deepfake audio detection literature. The paper critically examines the shortcomings of the existing approaches and positions its contribution in the ongoing research discourse. We chose MFCC features over newer self-supervised representations such as wav2vec 2.0 after carefully analyzing the trade-offs between computational constraints and performance, which demonstrates critical thinking about emerging issues in the field.
P1	The deepfake audio detection problem required deep technical knowledge in all aspects of audio signal processing, recurrent and convolutional neural network design, sequence modelling, data augmentation and binary classification. Each component was required to have a detailed theoretical understanding before practical implementation, confirming that the project

	operates at a high depth of knowledge consistent with advanced engineering practice.
P7	This is a truly interdisciplinary project, combining ideas from digital signal processing (MFCC extraction, sequence padding), machine learning (model training, dropout, optimization), cybersecurity (deepfake threat modeling, authentication systems), and software engineering (Python pipeline design, Google Colab experimentation). The project shows a strong interdependence of knowledge areas, as the correct integration of the domains results in a functional detection system.
A3	The project shows innovation by comparing three architectures systematically under strictly controlled conditions, providing reproducible empirical evidence not previously reported for this particular dataset. The custom amplitude-scaling augmentation strategy was a practical and novel choice of preprocessing for dealing with class imbalance without oversampling. The results challenge the assumption that more complex architectures always yield better results and offer actionable guidance for practitioners to select models for real-world audio authentication systems.

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